

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA0704 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I LAKSHITHA SREE K(192511336)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Calculation

File Size = **150 MB = 157,286,400 Bytes**

Number of Segments
 $= 157,286,400 \div 1500$

= **104,858 Segments**

Number of Frames
= **104,858 Frames**

Total Overhead
 $= 104,858 \times 40$
= **4,194,320 Bytes (≈ 4 MB)**

Transmission Time
 $= (157,286,400 \times 8) \div 50,000,000$
= **25.2 seconds**

Answer

- **Segments:** 104,858
- **Frames:** 104,858
- **Data Link Overhead:** 4,194,320 Bytes (≈ 4 MB)
- **Transmission Time:** 25.2 seconds

Theory

The **Transport Layer** divides the file into smaller segments and ensures reliable communication using sequencing, acknowledgements, and retransmissions. The **Data Link Layer** converts segments into frames by adding headers and trailers, performs error detection using CRC, and provides reliable communication between directly connected devices. These two layers together ensure accurate, efficient, and error-free data transmission across the network.

Question 2

Calculation

Window Size = **6 Frames**

Total Frames = **48**

Number of Windows

$$= 48 \div 6$$

= **8 Windows**

One frame is lost in every window.

Total Retransmissions

= **8 Frames**

Answer

- **Window Size:** 6 Frames
- **Windows Required:** 8
- **Retransmissions:** 8 Frames

Theory

The **Sliding Window Protocol** is a flow control mechanism used by the Transport Layer to improve communication efficiency. It allows the sender to transmit multiple frames before receiving an acknowledgment, reducing waiting time and increasing network throughput. If a frame is lost, only the missing frame is retransmitted instead of sending all frames again. This improves reliability, minimizes unnecessary retransmissions, and ensures efficient utilization of network bandwidth.

Question 3

Calculation

Packet Size = **12,000 Bytes**

MTU = **1500 Bytes**

Number of Fragments

$$= 12,000 \div 1500 + 1 \text{ (last fragment)}$$

= **9 Fragments**

IP Header Size = **20 Bytes**

Total Header Overhead

$$= 9 \times 20$$

= **180 Bytes**

Answer

- **Fragments Created:** 9
- **IP Header Overhead:** 180 Bytes

Theory

The **Network Layer** performs fragmentation when the packet size exceeds the Maximum Transmission Unit (MTU) of a network. Each fragment is assigned an IP header containing identification and fragment offset information. At the destination, all fragments are reassembled to reconstruct the original packet. This process enables successful transmission across networks with different MTU sizes.

Question 4

Calculation

Window Size = **6 Frames**

Total Frames = **48**

Number of Windows

$$= 48 \div 6$$

= **8 Windows**

One frame lost in each window

Retransmissions

= **8 Frames**

Answer

- **Windows Required: 8**
- **Retransmissions: 8 Frames**

Theory

Flow control regulates the transmission rate between the sender and receiver to avoid buffer overflow. It ensures that the sender transmits data only at a speed that the receiver can process. This improves communication efficiency, reduces congestion, and minimizes unnecessary retransmissions, leading to reliable data transfer.

Question 5

Calculation

File Size = **600 MB**

Checkpoint Interval = **60 MB**

Failure occurs after **390 MB**

Completed Checkpoints

$$= 390 \div 60$$

= **6 Checkpoints**

Data after last checkpoint

$$= 390 - 360$$

= **30 MB**

Data to be Retransmitted

= **30 MB**

Answer

- **Completed Checkpoints: 6**
- **Data Retransmitted: 30 MB**

Theory

The **Session Layer** establishes, manages, and terminates communication sessions between devices. It uses synchronization checkpoints during large file transfers so that communication can resume from the last saved checkpoint if a failure occurs. This reduces retransmission time, saves bandwidth, and improves the reliability of data communication.

Question 6

Calculation

Original File Size = **900 MB**

Compression = **40%**

Compressed Size

$$= 900 - (40\% \text{ of } 900)$$

$$= 900 - 360$$

= **540 MB**

Encryption Overhead = 5%

Encrypted Size

= $540 + (5\% \text{ of } 540)$

= $540 + 27$

= **567 MB**

Overall Size Reduction

= $((900 - 567) \div 900) \times 100$

= **37%**

Answer

- **Compressed Size:** 540 MB
- **Encrypted Size:** 567 MB
- **Overall Reduction:** 37%

Theory

The **Presentation Layer** is responsible for data formatting, compression, encryption, and decryption. Compression reduces the amount of data transmitted, improving bandwidth utilization and transmission speed. Encryption protects the data from unauthorized access during transmission, ensuring confidentiality and secure communication between sender and receiver.

Question 7

Calculation

File Size = **3 GB**

1 GB = **1024 MB**

Total File Size

= 3×1024

= **3072 MB**

Request Size = **3 MB**

Number of Requests

= $3072 \div 3$

= **1024 Requests**

Bandwidth = **120 Mbps**

Transmission Time

= $(3072 \times 8) \div 120$

= **204.8 Seconds**

Answer

- **Requests Generated:** 1024
- **Transmission Time:** 204.8 Seconds

Theory

The **Application Layer** provides services directly to end users through protocols such as HTTP, FTP, SMTP, and DNS. It manages file transfers, web browsing, email communication, and remote access. In FTP communication, files are divided into requests and transferred efficiently while maintaining reliable communication between client and server.

Question 8

Calculation

Application Layer Delay = **5 ms**

Transport Layer Delay = **8 ms**

Network Layer Delay = **12 ms**

Data Link Layer Delay = **10 ms**

Physical Layer Delay = **10 ms**

Total End-to-End Delay

= $5 + 8 + 12 + 10 + 10$

= **45 ms**

Answer

- **Total End-to-End Delay:** 45 ms

Theory

End-to-end delay is the total time taken for data to travel from the sender to the receiver. It includes processing delay, transmission delay, propagation delay, and queuing delay across different OSI layers. Minimizing these delays improves network performance, increases communication speed, and enhances the overall Quality of Service (QoS).

Question 9

Calculation

File Size = **150 MB**

Frame Size = **3000 Bytes**

Total Frames

= $157,286,400 \div 3000$

≈ **52,429 Frames**

Frame Overhead = **80 Bytes**

Total Overhead

= $52,429 \times 80$

= **4,194,320 Bytes**

Transmission Efficiency

= $(150 \div 157.286) \times 100$

≈ **95.24%**

Answer

- **Frames Required:** 52,429
- **Protocol Overhead:** 4,194,320 Bytes
- **Transmission Efficiency:** 95.24%

Theory

Protocol overhead refers to the additional information added by networking protocols for addressing, synchronization, and error detection. Although it slightly reduces bandwidth efficiency, protocol overhead is essential for reliable communication. The Data Link Layer uses this information to ensure error-free frame transmission between devices.

Question 10

Calculation

Original File Size = **240 MB**

Compression = **30%**

Compressed Size

= $240 - (30\% \text{ of } 240)$

= **168 MB**

Segment Size = **1400 Bytes**

Total Segments

= $176,160,768 \div 1400$

$\approx$ **125,830 Segments**

Protocol Overhead

= $125,830 \times 60$

= **7,549,800 Bytes**

Total Data Transmitted

= $168 \text{ MB} + 7.2 \text{ MB}$

$\approx$ **175.55 MB**

Answer

- **Compressed File Size:** 168 MB
- **Segments Created:** 125,830
- **Protocol Overhead:** 7,549,800 Bytes
- **Total Data Transmitted:** 175.55 MB

Theory

During data transmission, multiple OSI layers work together to ensure efficient communication. The Presentation Layer compresses data to reduce file size, the Transport Layer divides the data into segments, and the Data Link Layer adds headers and trailers for framing and error detection. This layered approach improves reliability, security, and overall network performance while ensuring successful end-to-end communication.